

Bit-Limited Sub-Nyquist Pulse-Doppler Radar

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Pulse-Doppler (PD) radars, which prefer to operate with large bandwidth signals to attain superior range resolution, are widely used due to their superior performance in target detection and parameter estimation. However, the increasing data flow of such radar systems induced by large bandwidth presents significant challenges to signal processing and hardware implementation, particularly concerning analog-to-digital converters (ADCs), which are tasked with converting high-bandwidth inputs into digital representations at or above the Nyquist rate. In this work, we introduce a hybrid analog-digital processing architecture, which implements sub-Nyquist sampling as well as low-bit quantization, specifically tailored for PD radar applications. We refer to the designed architecture as the bit-limited sub-Nyquist pulse-Doppler radar (BiLiPD) receiver, designed with cost-efficiency in mind, utilizing low-rate and low-resolution ADCs to reduce both cost and power consumption. Specifically, the received radar echoes are acquired under the sub-Nyquist sampling framework, and then, quantized with low-resolution ADCs. The building modules of the proposed BiLiPD receiver, including the analog preprocessing, the ADCs, and the digital processing, are jointly designed to mitigate the challenges posed by sub-Nyquist sampling and low-bit quantization. We incorporate structural constraints into the design problem, formulating it under the task-based quantization framework and employing gradient descent methods for its solution. Our simulation results illustrate that the proposed BiLiPD receiver operating under low-rate low-bit constraints is capable of accurately recovering target parameters, with its target recovery performance approaching that of classic PD processing operating with unlimited resolution ADCs

Received 8 November 2023; revised 25 March 2024 and 5 August 2024; accepted 23 September 2024. Date of publication 27 September 2024; date of current version 14 April 2025.

DOI: No. 10.1109/TAES.2024.3469174

Refereeing of this contribution was handled by F. Gini.

This work was supported in part by the Natural Science Foundation of Jiangsu Province, China, under Grant BK20221486, and in part by the National Natural Science Foundation of China under Grant 62471230.

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while notably outperforming that employing task-ignorant low-bit quantization.

I. INTRODUCTION

Pulse-Doppler (PD) radars [1] are widely applied in target detection, localization, and classification. They prefer to operate with large bandwidths for high range resolution, enabling high-resolution target detection and parameter estimation. In a typical radar receiver, the received analog signals are converted to digital representations before digital signal processing by using a set of analog-to-digital converters (ADCs). However, converting these signals with large bandwidth requires high-rate ADCs, resulting in large-scale dataflow and heavy burdens on hardware implementation. These challenges significantly increase the complexity of system design and signal processing, limiting the use of radar systems with power consumption and hardware implementation constraints. Consequently, there is a need for emerging schemes that can operate effectively with limited power and hardware implementation.

To break the bridge between the sampling rate and the signal bandwidth, compressive sensing (CS) [2] has been proposed to cope with the high-rate sampling demands. Specifically, it becomes possible to sample a signal below the Nyquist sampling rate, thus greatly alleviating the burden on hardware implementation and signal processing. Built upon the CS theory, various sub-Nyquist sampling architectures [3], [4], [5] have emerged to reduce the sampling rate significantly. In the realm of radar applications, the sub-Nyquist sampling can be performed in the time domain [6], [7], Doppler domain [8], spatial domain [9], or simultaneously multiple domains [10], [11], while maintaining high-resolution performance. Interested readers are referred to a survey paper [12], covering both theoretical and practical aspects of sub-Nyquist radars. Nevertheless, these works only focus on reducing the sampling rate while ignoring the effect of quantization on the sampled signals, assuming infinite precision quantization after sampling. In fact, high-resolution quantizers tend to be costly and power-hungry since the energy consumption of ADCs typically increases exponentially with the effective number of bits and linearly with the sampling rate [13], [14]. The related research of power saving issue of low-resolution ADCs is detailed in [15] and [16].

Recently designing radar systems with low-resolution ADCs has received great attention due to their low complexity and energy efficiency. One way to design such systems is to acquire the analog signals using low-bit quantizers, and then, design the digital processing to compensate for the induced quantization distortion. Most of the current works assume the use of one-bit sampling and focus on developing sophisticated algorithms to detect targets from one-bit measurements. For example, signal processing methodologies for PD radars and multiple-input and multiple-output (MIMO) radars with one-bit sampling have been developed in [17], [18], [19], [20], [21], and [22], respectively. Wu et al. [23] extend the works of one-bit quantization to

few-bit quantization and develop the corresponding signal processing. Although digital processing algorithms have been developed to mitigate the distortion resulting from low-resolution quantization, the overall performance of these systems is still significantly impacted since the quantizers are ignorant of the system tasks.

An alternative approach to designing a low-bit quantized system is incorporating the underlying task into the quantizer design [24]. By doing so, the quantizer can be optimized to perform better on the specific task at hand, leading to improved system performance. Specifically, the task-based quantization scheme comprises analog preprocessing, low-resolution ADCs, and digital processing, which results in a **hybrid analog–digital (HAD)** architecture. Such HAD architectures can be optimized by considering the task of extracting the desired information from the acquired analog signals. In radar systems, the task is to detect targets or resolve the target parameters rather than obtaining accurate digital representations of the received signals as in conventional quantization schemes. It has been shown in MIMO radar [25], [26] and MIMO communication systems [24], [27], [28], [29] that the task-based quantization scheme can significantly outperform the task-ignorant quantization scheme when operating with limited bit budgets. These works mainly focus on MIMO systems with multiple input channels, which can be optimized as HAD models directly following the task-based quantization methods.

In this work, we focus on the development of a low-bit quantized sub-Nyquist sampling architecture for PD radar systems. An HAD processing architecture, which seamlessly integrates task-based quantization methods into sub-Nyquist PD radar processing, is proposed. Specifically, our study delves into the intricacies of designing a bit-limited acquisition system for PD radar within the sub-Nyquist sampling framework. We refer to the developed architecture as bit-limited sub-Nyquist pulse-Doppler (BiLiPD) receiver. To facilitate the design of this acquisition system, we implement a multichannel sub-Nyquist sampling architecture during the analog preprocessing phase. This architecture not only significantly narrows down the analog bandwidth entering the acquisition system but also presents a feasible analog processing for task-based quantization. Furthermore, due to the complex nonlinear relation between the targets' parameters and the radar echoes, it is difficult to directly formulate the target recovery as the task within the framework of task-based quantization. To overcome this challenge, we decompose the digital component of the BiLiPD receiver into two distinct parts. First, a digital filter is employed to acquire compressed representations of input signals. This digital filter can be jointly adjusted alongside the analog component and the ADC parameters, guided by the objective of minimizing the mean-square error (MSE) between the recovered compressed representations and the optimal linear estimates derived from unquantized data. The second part is dedicated to retrieving target parameters from the output of the digital filter, namely, the compressed representations, by employing the CS-based sparse recovery algorithm.

Our main contributions can be summarized as follows.

- 1) We introduce an HAD architecture by integrating task-based quantization with sub-Nyquist sampling in the context of PD radar processing. This architecture consisting of analog preprocessing, low-rate and low-resolution ADCs, and digital processing, enables the optimization design of a low-bit quantized sub-Nyquist radar receiver.
- 2) We optimize the BiLiPD receiver to reduce the effect of quantization distortion due to low-resolution ADCs while ensuring the sparse recovery performance within the sub-Nyquist sampling framework. By considering the structural constraint of analog preprocessing, the joint design of analog preprocessing, the quantizers, and the digital processing is formulated as a constrained optimization problem.
- 3) We present two designs of the BiLiPD receiver: The first is a special case that each processing channel outputs only one sample during each pulse repetition interval (PRI). In this case, the analog preprocessing matrix is block-diagonal. The design problem can be solved by optimizing each block iteratively. Then, we design the BiLiPD receiver for the general case where the analog preprocessing matrix is block-diagonal and Toeplitz. In this case, we first relax the Toeplitz structural constraint to find an optimal solution, and then, project it onto a Toeplitz matrix.
- 4) Numerical simulations are provided to demonstrate the performance of the BiLiPD receiver. The results show that the proposed receiver is capable of accurately recovering target parameters with its performance approaching that of classic PD processing without quantization, while significantly outperforming that with task-ignorant low-bit quantization.

The rest of this article is organized as follows. This article begins by introducing the system model of PD radar with HAD receiver architectures in Section II. It will then go on to present the architecture of the proposed BiLiPD receiver operating with cost-efficient ADCs in Section III. Section IV and V provide the theoretical analysis of BiLiPD receiver design method and numerical simulations, respectively. Finally, Section VI concludes this article.

Throughout this article, lower case (upper case) bold characters are used to denote vectors (matrices). $(\mathbf{x})_i$ and $(\mathbf{X})_{i,j}$ refer to the i th element of a vector $\mathbf{x}$ and the (i, j) th element of a matrix $\mathbf{X}$, respectively. $\mathbb{R}$ and $\mathbb{C}$ are used to denote the sets of real and complex numbers, respectively. $\mathbb{Z}$ and $\mathbb{N}$ are used to denote the sets of integers and positive integers, respectively. $(\cdot)^T$, $(\cdot)^H$, $\text{Tr}(\cdot)$, and $\text{sign}(\cdot)$ denote the matrix transposition, Hermitian transposition, trace, and sign operator, respectively. The term $\mathbf{I}_N$ is used to refer to the $N \times N$ identity matrix. Finally, $\lceil \cdot \rceil$ and $\lfloor \cdot \rfloor$ denote the ceiling and the floor functions, $a^+ \triangleq \max(a, 0)$, and $\text{vec}(\cdot)$ is the vectorization operator.

II. PD RADAR MODEL WITH SUB-NYQUIST SAMPLING

In this section, we introduce the signal model, the sub-Nyquist sampling architecture and the corresponding signal processing of a typical sub-Nyquist PD radar while ignoring the effect of distortion induced by quantization.

A. PD Radar Signal Model

A typical PD radar transmits Q pulses in a coherent processing interval (CPI), given by

$$s(t) = \sum_{q=0}^{Q-1} h(t - qT) e^{j2\pi f_c t}, 0 \leq t \leq QT \quad (1)$$

where T is the PRI, f_c is the carrier frequency, and $h(t)$ is a known pulsed signal with its pulsewidth T_p ($T_p \ll T$). We assume that the continuous-time Fourier transform $H(f) \triangleq \int_{-\infty}^{+\infty} h(t) e^{-j2\pi f t} dt$ has negligible energy at frequencies beyond $B/2$, and thus, we refer to B as the bandwidth of $h(t)$.¹

Supposed that there are K targets lying in the radar unambiguous time–frequency region, characterized by the distance R_k between the target and the radar receiver, the radial velocity v_k , and the complex reflection coefficient $\tilde{\alpha}_k$ for the k th target. Furthermore, we make the following assumptions on the targets.

- 1) *Swerling 1 model* [30]: For the Swerling 1 target model, the targets are supposed to appear in the far field with the radar cross-section being constant during a CPI consisting of multiple pulses.
- 2) *Slow targets*: The velocities of the moving targets are small that the induced time delays are nearly constant during a CPI and the induced Doppler shifts are assumed to be constant during a PRI.
- 3) *Small acceleration*: The accelerations of the targets are small that the targets' velocities are nearly constant during a CPI.
- 4) *Unambiguous targets*: The targets are within the unambiguous range and Doppler velocities of the radar.

Therefore, the received signal at the radar receiver is expressed as

$$\begin{aligned} \tilde{x}(t) &= \sum_{k=1}^K \tilde{\alpha}_k s(t - \tau_k) e^{j2\pi f_k t} + \tilde{n}(t) \\ &= \sum_{k=1}^K \sum_{q=0}^{Q-1} \tilde{\alpha}_k h(t - qT - \tau_k) e^{j2\pi f_c(t - \tau_k)} e^{j2\pi f_k t} + \tilde{n}(t) \end{aligned} \quad (2)$$

where $\tau_k = \frac{2R_k}{c}$ and $f_k = \frac{2v_k f_c}{c}$ denote the time delay and the Doppler frequency of the k th target, respectively. $\tilde{n}(t)$ denotes noise signal.

¹The term “bandwidth” (B) defined here differs from the traditional -4 dB bandwidth commonly referenced in radar terminology, which delineates the bandwidth where the signal power falls below -4 dB of the peak power. This definition is chosen for its analytical tractability.

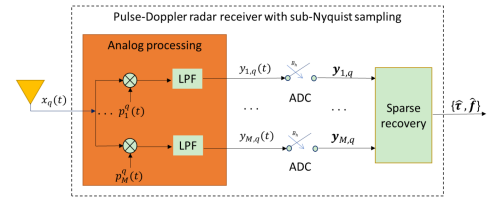


Fig. 1. Architecture of sub-Nyquist PD radar receiver.

After demodulation, the baseband component of the received signal, defined as $x(t) = \tilde{x}(t) e^{-j2\pi f_c t}$, can be denoted as

$$\begin{aligned} x(t) &\triangleq \sum_{k=1}^K \sum_{q=0}^{Q-1} \alpha_k h(t - qT - \tau_k) e^{j2\pi f_k t} + n(t) \\ &\approx \sum_{q=0}^{Q-1} x_q(t - qT) + n(t) \end{aligned} \quad (3)$$

where $\alpha_k = \tilde{\alpha}_k e^{-j2\pi f_c \tau_k}$, $n(t) = \tilde{n}(t) e^{-j2\pi f_c t}$, and $x_q(t) \triangleq \sum_{k=1}^K \alpha_k h(t - \tau_k) e^{j2\pi f_k qT}$ for $0 \leq t < T$ denotes the noise-free baseband component during the q th pulse. The second line of (3) follows the slow target assumption.

The goal of the PD radar is to resolve the K delay-Doppler pairs $\{\tau_k, f_k\}_{k=1}^K$, along with the corresponding K target reflection coefficients $\{\alpha_k\}$ from the received signal $x(t)$. Classic PD radars sample $x(t)$ at or above the Nyquist rate using high-resolution ADCs. Following this, digital matched filters are applied to detect the K targets on the delay-Doppler grids and retrieve the target reflection coefficients. To enhance both range and Doppler resolution, radar systems often employ signals with larger bandwidth and extend the duration of one CPI. However, this necessitates higher sampling rates in hardware design and generates a substantial volume of data for storage, transmission, and processing.

B. Sub-Nyquist Sampling Architecture

To overcome the limitations of high sampling rates imposed by PD radar systems while maintaining signal bandwidth, several sub-Nyquist sampling architectures have been developed, leveraging the sparsity of the received signal. Various analog preprocessing has been designed to ensure that the resulting sub-Nyquist sampling architecture satisfies the theoretical signal recovery conditions, such as the **restricted isometry property (RIP)** and coherence condition [31]. Using such sub-Nyquist sampling architectures, sparse recovery-based processing can be effectively applied to reconstruct the original signal or extract the desired information from the acquired sub-Nyquist samples.

In this article, we consider a multichannel sub-Nyquist sampling architecture, illustrated in Fig. 1, which is built upon the Xampling architecture in [32]. In this setup, the received signals in each of the M channels undergo a two-step process before being sampled. First, during the q th pulse, the m th channel multiplies the received signal $x_q(t)$ with a periodic function $p_m^q(t)$, which has the Fourier expansion,

denoted as a weighted sum of complex sinusoids

$$p_m^q(t) = \sum_{n=\lceil -B_q T/2 \rceil}^{\lfloor B_q T/2 \rfloor} \beta_{m,n}^q e^{j2\pi n t/T}, t \in [0, T) \quad (4)$$

where $\beta_{m,n}^q$ denotes the weight corresponding to the complex sinusoid $e^{j2\pi n t/T}$ during the q th pulse, and B_q denotes the bandwidth of $p_m^q(t)$. After this mixing operation, the output signal is passed through an ideal low-pass filter with bandwidth B_0 ($B_0 < B$). This filtering results in a bandwidth-reduced signal $y_{m,q}(t)$, which can be sampled at a rate below the Nyquist rate of the original signal. The mixing operation scrambles the spectrum of $x_q(t)$, such that a portion of the energy of all bands appears in baseband, ensuring that the low-pass-filtered signal $y_{m,q}(t)$ captures each of the spectrum components. This makes it feasible to recover the original signal. For the sake of discussion, we assume $B_q \geq B + B_0$ in the following sections to simplify our analysis.

Now let us derive the mathematical representation of the signal $y_{m,q}(t)$. For clarity and to facilitate our discussion, the analyses of the noise-free signal and noise components are conducted separately in the subsequent sections.

Since the baseband pure signal $x_q(t)$ is limited to $t \in [0, T)$, it can be represented by its Fourier series

$$x_q(t) = \sum_{n \in \mathbb{Z}} X_q[n] e^{j2\pi n t/T}, t \in [0, T) \quad (5)$$

where for $-\frac{BT}{2} \leq n \leq \frac{BT}{2} - 1$

$$\begin{aligned} X_q[n] &= \frac{1}{T} \int_0^T x_q(t) e^{-j2\pi n t/T} dt \\ &= \frac{1}{T} \sum_{k=1}^K \alpha_k e^{j2\pi f_k q T} e^{-j2\pi n \tau_k/T} H\left(\frac{2\pi n}{T}\right), \end{aligned} \quad (6)$$

Let $y_{m,q}^\dagger(t)$ be the noise-free version of $y_{m,q}(t)$. $y_{m,q}^\dagger(t)$ can be also represented by its Fourier series

$$y_{m,q}^\dagger(t) = \mathcal{LP}\{x_q(t)p_m^q(t)\} = \sum_{n=\lceil -B_0 T/2 \rceil}^{\lfloor B_0 T/2 \rfloor} \tilde{c}_{m,q}[n] e^{j2\pi n t/T} \quad (7)$$

where $\mathcal{LP}\{\cdot\}$ denotes the ideal low-pass filtering operation, and $\tilde{c}_{m,q}[n]$ denotes the Fourier coefficient of $x_q(t)p_m^q(t)$, i.e.,

$$\begin{aligned} \tilde{c}_{m,q}[n] &\triangleq \frac{1}{T} \int_0^T x_q(t) p_m^q(t) e^{-j2\pi n t/T} dt \\ &= \frac{1}{T} \sum_{n'=\lceil -BT/2 \rceil}^{\lfloor BT/2 \rfloor} \beta_{m,n-n'}^q H\left(\frac{2\pi n'}{T}\right) \\ &\quad \sum_{k=1}^K \alpha_k e^{j2\pi f_k q T} e^{-j2\pi n' \tau_k/T}. \end{aligned} \quad (8)$$

Here, the last line is derived with the assumption that $H(\frac{2\pi n}{T})$ has negligible energy for $|n| > \lfloor \frac{BT}{2} \rfloor$.

Let us define $N \triangleq BT$ and $N_0 \triangleq B_0 T$. We here assume N and N_0 to be odd in the following discussion for the sake of convenience. Let $c_q[n] \triangleq \sum_{k=1}^K \alpha_k e^{j2\pi f_k q T} e^{-j2\pi n \tau_k/T}$, $\mathbf{c}_q \triangleq [c_q[\frac{N-1}{2}], \dots, c_q[\frac{N-1}{2}]]^T \in \mathbb{C}^N$, and $\tilde{\mathbf{c}}_{m,q} \triangleq [\tilde{c}_{m,q}[\frac{N_0-1}{2}], \dots, \tilde{c}_{m,q}[\frac{N_0-1}{2}]]^T \in \mathbb{C}^{N_0}$. Then, (8) can be written as the following matrix form:

$$\tilde{\mathbf{c}}_{m,q} = \mathbf{B}_{m,q} \mathbf{H} \mathbf{c}_q \quad (9)$$

where $\mathbf{H} = \text{diag}(H(\frac{-\pi(N-1)}{T}), \dots, H(\frac{\pi(N-1)}{T})) \in \mathbb{C}^{N \times N}$ is a diagonal matrix representing the spectrum components of the transmitted signal $h(t)$, $\mathbf{B}_{m,q} \in \mathbb{C}^{N_0 \times N}$ is a matrix determined by the weights of the mixing complex sinusoids $p_m^q(t)$, i.e., $\mathbf{B}_{m,q} =$

$$\begin{bmatrix} \beta_{m, \frac{N-N_0}{2}}^q & \beta_{m, \frac{N-N_0}{2}-1}^q & \cdots & \beta_{m, -\frac{N+N_0}{2}+1}^q \\ \beta_{m, \frac{N-N_0}{2}+1}^q & \beta_{m, \frac{N-N_0}{2}}^q & \cdots & \beta_{m, -\frac{N+N_0}{2}+2}^q \\ \vdots & \ddots & \ddots & \vdots \\ \beta_{m, \frac{N+N_0}{2}-1}^q & \cdots & \cdots & \beta_{m, \frac{N_0-N}{2}}^q \end{bmatrix} \quad (10)$$

which is a Toeplitz matrix. Inserting the expression (9) into (7), we have

$$y_{m,q}^\dagger(t) = \mathbf{e}^T(t) \mathbf{B}_{m,q} \mathbf{H} \mathbf{c}_q \quad (11)$$

where $\mathbf{e}(t) = [e^{-j\pi(N_0-1)t/T}, \dots, e^{j\pi(N_0-1)t/T}]^T$.

We now express the noise-free output of the M analog processing channels during the q th pulse in multivariate form by defining $\mathbf{y}_q^\dagger(t) \triangleq [y_{1,q}^\dagger(t), y_{2,q}^\dagger(t), \dots, y_{M,q}^\dagger(t)]^T$, and obtain from (11) that

$$\mathbf{y}_q^\dagger(t) = \mathbf{E}(t) \mathbf{B}_q \mathbf{H} \mathbf{c}_q \quad (12)$$

where $\mathbf{E}(t) = \mathbf{I}_M \otimes \mathbf{e}^T(t) \in \mathbb{C}^{M \times MN_0}$, and $\mathbf{B}_q = [\mathbf{B}_{1,q}^T, \dots, \mathbf{B}_{M,q}^T]^T \in \mathbb{C}^{MN_0 \times N}$. Using the similar derivation, the noise signal during the q th pulse after analog processing can be expressed as

$$\mathbf{w}_q(t) = \mathbf{E}(t) \mathbf{B}_q \mathbf{n}_q \quad (13)$$

where $\mathbf{n}_q \in \mathbb{C}^N$ is the frequency-domain representation of the noise signal during the q th pulse.

After analog preprocessing, the M noisy signals $\mathbf{y}_q(t) \triangleq \mathbf{y}_q^\dagger(t) + \mathbf{w}_q(t)$ are forwarded to the ADCs, where the analog signals are converted into digital streams. The conversion is done with low-rate ADCs instead of costly high-rate ADCs due to reducing the signal's bandwidth. The overall sampling rate now is MB_0 , which achieves the sub-Nyquist sampling if B_0 is less than B/M , i.e., $B_0 < B/M$.

Let $\mathbf{y}_q[i] \triangleq \mathbf{y}_q(\frac{i}{B_0})$, $i = 0, 1, \dots, N_0 - 1$, be the samples of $\mathbf{y}_q(t)$ with the sampling interval $\frac{1}{B_0}$. Define the $MN_0 \times 1$ sampling vector $\mathbf{y}_q = [\mathbf{y}_q^T[0], \dots, \mathbf{y}_q^T[N_0 - 1]]^T$. Then, by using the expressions (12) and (13), we have

$$\mathbf{y}_q = \mathbf{y}_q^\dagger + \mathbf{w}_q = \mathbf{E} \mathbf{B}_q \mathbf{H} \mathbf{c}_q + \mathbf{E} \mathbf{B}_q \mathbf{n}_q \quad (14)$$

where $\mathbf{E} = [\mathbf{E}^T(0), \dots, \mathbf{E}^T(\frac{N_0-1}{B_0})]^T \in \mathbb{C}^{MN_0 \times MN_0}$.

Due to the sub-Nyquist sampling, classic PD processing is no longer applicable here. We will briefly introduce the sub-Nyquist PD processing in the next subsection.

C. Sub-Nyquist PD Processing

The aim of sub-Nyquist PD processing is to recover the delay-Doppler pairs $\{\tau_k, f_k\}_{k=1}^K$ as well as the target reflection coefficients $\{\alpha_k\}$ from the sub-Nyquist data $\{\mathbf{y}_q\}_{q=0}^{Q-1}$. For convenience, let us assume that the parameters τ_k and f_k are located on the Nyquist grid, i.e., $\tau_k \in \{\frac{Tl}{N}\}_{l=0}^{N-1}$ and $f_k \in \{-\frac{1}{2T} + \frac{l}{TQ}\}_{l=0}^{Q-1}$. According to the definition of $c_q[n]$, the matrix $\mathbf{C} = [\mathbf{c}_0, \dots, \mathbf{c}_{Q-1}] \in \mathbb{C}^{N \times Q}$ obeys the following sparse representation:

$$\mathbf{C} = \mathbf{U}\mathbf{A}\mathbf{V}^T \quad (15)$$

where $\mathbf{U} \in \mathbb{C}^{N \times N}$ with $(\mathbf{U})_{n,l} = e^{-j2\pi nl/N}$, $\mathbf{V} \in \mathbb{C}^{Q \times Q}$ with $(\mathbf{V})_{q,l} = e^{j2\pi q(-1/2+1/Q)}$, and $\mathbf{A} \in \mathbb{C}^{N \times Q}$ is a sparse matrix that contains K nonzero elements corresponding to the target reflection coefficients. The sparsity pattern of $\mathbf{A}$ encapsulates the information of target parameters, i.e., if the k th nonzero element of $\mathbf{A}$ is located at its index (l_1, l_2) where $0 \leq l_1 < N$ and $0 \leq l_2 < Q$, then we can get the time delay and Doppler frequency of the k th target, denoting as $\tau_k = \frac{Tl_1}{N}$ and $f_k = -\frac{1}{2T} + \frac{l_2}{TQ}$.

Let $\mathbf{c} = \text{vec}(\mathbf{C})$, we represent $\mathbf{c}$ as

$$\mathbf{c} = \Phi \mathbf{a} \quad (16)$$

where $\mathbf{a} = \text{vec}(\mathbf{A}) \in \mathbb{C}^{NQ}$ is a K -sparse vector, and $\Phi = \mathbf{V} \otimes \mathbf{U} \in \mathbb{C}^{NQ \times NQ}$. Using the sparse representation in (16), the sampled output $\mathbf{y} = [\mathbf{y}_0^T, \mathbf{y}_1^T, \dots, \mathbf{y}_{Q-1}^T]^T \in \mathbb{C}^{MN_0Q}$ becomes

$$\mathbf{y} = \bar{\mathbf{E}}\bar{\mathbf{B}}(\bar{\mathbf{H}}\Phi\mathbf{a} + \mathbf{w}) \quad (17)$$

with $\mathbf{w} = [\mathbf{w}_0^T, \mathbf{w}_1^T, \dots, \mathbf{w}_{Q-1}^T]^T \in \mathbb{C}^{MN_0Q}$, $\bar{\mathbf{E}} = \mathbf{I}_Q \otimes \mathbf{E} \in \mathbb{C}^{MN_0Q \times MN_0Q}$, $\bar{\mathbf{H}} = \mathbf{I}_Q \otimes \mathbf{H} \in \mathbb{C}^{NQ \times NQ}$, and $\bar{\mathbf{B}} = \text{blkdiag}(\mathbf{B}_0, \dots, \mathbf{B}_{Q-1}) \in \mathbb{C}^{MN_0Q \times NQ}$.

By exploiting the sparsity of the received signal, the reconstruction of the delay-Doppler pairs $\{\tau_k, f_k\}_{k=1}^K$ can be formulated as a sparse recovery problem by solving the following ℓ_1 minimization:

$$\hat{\mathbf{a}} = \min_{\mathbf{a}} \|\mathbf{a}\|_1 \quad \text{s.t.} \quad \|\mathbf{y} - \Psi\mathbf{a}\|_2^2 \leq \varepsilon \quad (18)$$

or the least absolute shrinkage and selection operator (LASSO) problem

$$\hat{\mathbf{a}} = \min_{\mathbf{a}} \|\mathbf{y} - \Psi\mathbf{a}\|_2^2 + \rho \|\mathbf{a}\|_1 \quad (19)$$

where $\Psi \triangleq \bar{\mathbf{E}}\bar{\mathbf{B}}\bar{\mathbf{H}}\Phi \in \mathbb{C}^{MN_0Q \times NQ}$ denotes the equivalent CS measurement matrix, ε and ρ are predefined parameters. The optimization problems in (18) and (19) can be solved by greedy algorithms, such as orthogonal matching pursuit (OMP) [33], or convex optimization methods, such as fast iterative shrinkage thresholding algorithm (FISTA) [34]. According to the works in [2], the equivalent CS measurement matrix Ψ satisfies the RIP condition with high probability when using random mixing signals. Thus, the error bound of the target reconstruction performance can be bounded, as shown in [2].

However, in the current sub-Nyquist sampling architecture, the quantization distortion is commonly neglected by assuming the use of high-resolution ADCs. In practical

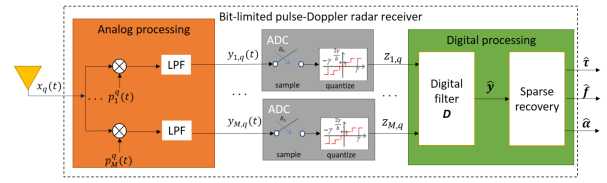


Fig. 2. Task-based quantization for PD radar.

scenarios, the quantization distortion is inevitable. Moreover, due to the considerable hardware complexity associated with high-resolution ADCs, there is a growing tendency to use low-resolution ADCs, resulting in significant quantization distortion. Designing the sub-Nyquist PD processing while explicitly considering the quantization distortion thus presents a formidable challenge.

III. BILIPD RECEIVER ARCHITECTURE

This section presents the proposed BiLiPD receiver, built upon the sub-Nyquist sampling architecture introduced in Section II-B. Specifically, Section III-A introduces the bit-limited pulse-Doppler radar (BiLiPD) architecture and models the effect of quantization, then Section III-B formally formulates the design problem by explicitly taking account of the quantization distortion.

A. BiLiPD Architecture

The proposed BiLiPD receiver is designed to facilitate the PD processing with the bit budget constraints. As depicted in Fig. 2, the BiLiPD receiver shares the same architecture as the sub-Nyquist sampling architecture in Fig. 1, with one crucial distinction. Instead of employing high-resolution ADCs, the BiLiPD receiver uses low-resolution ADCs. Thus, the effect of the induced quantization distortion has to be taken into account.

Specifically, in the proposed BiLiPD receiver, we assume that the sampling rate of the BiLiPD receiver is the same as that of the sub-Nyquist architecture introduced in Section II-B and M uniform complex-valued quantizers with decision regions b and support $\gamma > 0$ are employed to convert the signals $\mathbf{y}_q(t)$ into digital representations. These complex-valued quantizers independently discretize the real and imaginary parts of each analog input sample and the resulting quantization mapping is given as $\mathcal{Q}_C^{\gamma,b}(\cdot) = \mathcal{Q}^{\gamma,b}(\cdot) + j\mathcal{Q}^{\gamma,b}(\cdot)$, where $\mathcal{Q}^{\gamma,b}(\cdot)$ is the real-valued quantization operator applied element-wise to any real vector or matrix, given by

$$\mathcal{Q}^{\gamma,b}(x) = \begin{cases} -\gamma + \frac{2\gamma}{b}(l + \frac{1}{2}), & x - l\frac{2\gamma}{b} + \gamma \in [0, \frac{2\gamma}{b}] \\ \text{sign}(x)(\gamma - \frac{\gamma}{b}), & |x| > \gamma. \end{cases}$$

The quantized output, denoted as $\mathbf{z}_q \in \mathbb{C}^{MN_0}$, is thus given by

$$\mathbf{z}_q = \mathcal{Q}_C^{\gamma,b}(\mathbf{y}_q) = \mathcal{Q}_C^{\gamma,b}(\mathbf{E}\mathbf{B}_q(\mathbf{H}\mathbf{c}_q + \mathbf{w}_q)). \quad (20)$$

Using the sparse representation in (16), the quantized output $\mathbf{z} = [\mathbf{z}_0^T, \mathbf{z}_1^T, \dots, \mathbf{z}_{Q-1}^T]^T \in \mathbb{C}^{MN_0Q}$ becomes

$$\mathbf{z} = \mathcal{Q}_C^{\gamma, b}(\mathbf{y}) = \mathcal{Q}_C^{\gamma, b}(\bar{\mathbf{E}}\bar{\mathbf{B}}(\bar{\mathbf{H}}\Phi\mathbf{a} + \mathbf{w})) = \mathcal{Q}_C^{\gamma, b}(\Psi\mathbf{a} + \mathbf{w}^\dagger) \quad (21)$$

where $\Psi = \bar{\mathbf{E}}\bar{\mathbf{B}}\bar{\mathbf{H}}\Phi$, and $\mathbf{w}^\dagger = \bar{\mathbf{E}}\bar{\mathbf{B}}\mathbf{w}$. The total number of bits used to represent $\mathbf{z}$ is $MN_0Q\lceil\log b\rceil$. The quantized output $\mathbf{z}$ is then forwarded to the digital processing to resolve the target parameters. When faced with a limited bit budget, allocating a minimal number of bits to each measurement becomes essential. However, this can lead to significant quantization distortion, making the accurate resolution of target parameters a formidable challenge.

In our design, we model the quantizers as implementing non-subtractive uniform dithered quantization [35], leading to the fact that the resulting quantization noise is white and uncorrelated with the input. Although the dithered quantization facilitates the design and analysis, the proposed system can also be applied without dithering. Furthermore, our derivation is carried out with nonoverloaded quantizers that the magnitudes of the input signals lie within the support $[-\gamma, \gamma]$ of the quantizers with sufficiently large probability, so as to avoid bringing about additional distortion induced by quantization saturation. Here, we set γ to be some multiple η of the maximal standard derivation of the inputs, given by

$$\gamma^2 = \eta^2 \max_{l=1,2,\dots,MN_0Q} \mathbb{E}\{|\mathbf{y} + \boldsymbol{\omega}|_l^2\} \quad (22)$$

where $\boldsymbol{\omega}$ denotes the dither signal whose real and imaginary part are i.i.d. distributed uniformly within $[-\frac{\Delta}{2}, \frac{\Delta}{2}]$ with $\Delta = \frac{2\gamma}{b}$. According to Chebyshev's inequality, by setting $\eta^2 \geq 3$, the amplitude of the input is smaller than the dynamic range with probability over 89%. Thus, the tunable parameter of the quantizer is the support γ .

The aim of designing the BiLiPD receiver is to facilitate the recovery of the delay-Doppler pairs $\{\tau_k, f_k\}_{k=1}^K$ and the target reflection coefficients $\{\alpha_k\}$ from the quantized measurements $\mathbf{z}$. It is achieved through the joint optimization of its building modules, such as analog preprocessing, quantizers, and digital processing. Our approach resorts to the task-based quantization framework [24] to solve our problem, which is developed to recover the defined linear task rather than the original unquantized signal from the quantized data. Unfortunately, due to the intricate nonlinear relationship between the targets' parameters $\{\tau_k, f_k, \alpha_k\}_{k=1}^K$ and the received signal $\mathbf{y}$, it is difficult to directly define the targets' parameters as the task for recovery, as done in [24]. In the proposed BiLiPD receiver, we partition digital processing into two stages, as depicted in Fig. 2. The first stage employs a digital filter $\mathbf{D} \in \mathbb{C}^{p \times MN_0Q}$ to recover an intermediate task (IT) vector $\mathbf{s}$, where p denotes the output dimension of the digital filter. We assume linear digital filtering operations, i.e., the output of the digital filter is $\mathbf{D}\mathbf{z}$. The IT vector $\mathbf{s}$ is carefully chosen to enable the estimation of the targets' parameters $\{\tau_k, f_k, \alpha_k\}_{k=1}^K$ using the conventional methods mentioned in Section II-C. The parameter estimation will be performed in the second stage of digital processing. We will specify the optimization problem in the following subsection.

B. Optimization Problem Formulation

In this subsection, we formally formulate the design problem of the BiLiPD receiver. How to choose the appropriate IT vector and performance metrics to formulate our problem will be addressed.

Now we first explicitly define the IT vector to facilitate the joint design under the framework of task-based quantization. In our previous research [25], we used a surrogate measurement, obtained as a linear combination of the sparse vector $\mathbf{a}$, as the task vector. However, according to the sub-Nyquist PD processing in Section II-C, a more straightforward alternative is to choose the sub-Nyquist measurements $\mathbf{y}$ as the IT vector, i.e., $\mathbf{s} = \mathbf{y}$. Unlike the approach presented in [25], where the desired task vector remains independent of the measurement system, the task vector introduced here becomes a function of the analog processing matrix $\bar{\mathbf{B}}$. This dependence significantly elevates the complexity involved in the design of the BiLiPD system. In this case, the sparse recovery at the second stage of digital processing is equivalent to solve the problem (18), by replacing $\mathbf{y}$ with the output of the digital filter $\hat{\mathbf{y}} = \mathbf{D}\mathbf{z}$. Consequently, the performance of the sparse recovery hinges upon the properties of the equivalent CS measurement matrix Ψ . It follows that the design of the BiLiPD receiver necessitates a comprehensive consideration of both the effect of quantization distortion and the induced sparse recovery problem.

Before formally formulating our joint design problem of the BiLiPD receiver, we discuss the performance metrics used to evaluate the effects of quantization distortion as well as the sparse recovery, respectively, as follows.

- 1) *Effect of quantization distortion:* We rely on the task-based quantization framework [24] to deal with the induced quantization distortion. Under the task-based quantization framework, we jointly design the analog processing, the quantizers, and the digital filter of the BiLiPD receiver to minimize the MSE between the output of the digital filter $\hat{\mathbf{y}} = \mathbf{D}\mathbf{z}$ and the desired IT vector $\mathbf{y}$, which is denoted as $\epsilon_{\text{MSE}} = \mathbb{E}\{\|\hat{\mathbf{y}} - \mathbf{y}\|_2^2\}$. Noting that although our task is to minimize the ϵ_{MSE} , the true task of this acquisition system is to resolve the targets' parameters $\{\tau_k, f_k, \alpha_k\}_{k=1}^K$, i.e., to recover the sparse vector $\mathbf{a}$. According to the sparse recovery problem in (18), the approximation error between $\mathbf{y}$ and $\hat{\mathbf{y}}$ can be viewed as the additive noise term, which has a great impact on the recovery performance of the sparse vector $\mathbf{a}$ [2]. In other words, by minimizing ϵ_{MSE} , the BiLiPD receiver is optimized to the benefit of the recovery of sparse vector $\mathbf{a}$.
- 2) *Effect of sparse recovery:* In the sub-Nyquist sampling architecture considered in Section II-B, the resulted equivalent CS measurement matrix Ψ satisfies the RIP condition with high probability when using random analog processing matrix $\bar{\mathbf{B}}$ [2]. However, to minimize ϵ_{MSE} , the analog processing matrix $\bar{\mathbf{B}}$ becomes deterministic. Although the computation of the RIP constant is known to be NP-hard, it has been

realized that a low coherence between columns of Ψ is a desired property in the CS framework. The mutual coherence of the matrix Ψ can be defined as the maximum absolute value and normalized inner product between all columns, i.e.,

$$\mu(\Psi) = \max_{i \neq j} \frac{|\psi_i^T \psi_j|}{\|\psi_i\|_2 \|\psi_j\|_2} \quad (23)$$

where ψ_i denotes the i th column of Ψ . The mutual coherence is equivalent to compute the maximum absolute off-diagonal elements in the Gram matrix $\mathbf{G} = \tilde{\Psi}^H \tilde{\Psi}$, where $\tilde{\Psi}$ is a column-normalized version of Ψ . In other word, minimizing the mutual coherence is to find a measurement matrix Ψ making the corresponding Gram matrix as close to an identity matrix as possible. Therefore, to guarantee the performance of sparse recovery, we engage in minimizing the term $\|\Psi^H \Psi - \xi \mathbf{I}_{NQ}\|_F^2$, where ξ is a regularization parameter to take account of the energy of each column in Ψ . Optimizing this term avoids the need to compute the column-normalized version of Ψ and allows us to directly design the measurement matrix Ψ .

To sum up, our goal is to jointly design the analog processing matrix $\tilde{\mathbf{B}}$, the digital filter $\mathbf{D}$, and the support of the quantizer γ in the BiLiPD receiver to minimize the error ϵ_{MSE} as well as the mutual coherence of Ψ . The optimization problem is also restricted by the block diagonal structure of the matrix $\tilde{\mathbf{B}}$, where each block matrix is a Toeplitz matrix, as shown in Section II-B. Therefore, our design problem can be formulated as

$$\begin{aligned} \min_{\{\mathbf{B}_{m,q}\}, \mathbf{D}, \gamma} \quad & \epsilon_{\text{MSE}} + \lambda_C \|\Psi^H \Psi - \xi \mathbf{I}_{NQ}\|_F^2 \\ \text{s.t.} \quad & \tilde{\mathbf{B}} = \text{blkdiag}(\mathbf{B}_1, \dots, \mathbf{B}_Q) \\ & \mathbf{B}_{m,q} \in \mathbb{T}^{N_0 \times N}, m = 1, 2, \dots, M \\ & q = 0, 1, \dots, Q-1 \end{aligned} \quad (24)$$

where $\mathbb{T}^{N_0 \times N}$ denotes the set of all the $N_0 \times N$ Toeplitz matrices and $\lambda_C > 0$ is the regularization parameter balancing the two terms. In the following section, we will solve the optimization problem (24) following the framework of task-based quantization.

IV. OPTIMAL DESIGN OF THE BILIPD RECEIVER

In this section, we study the design of BiLiPD receiver with the task of target parameter estimation under the bit budget and hardware implementation constraints. Based on the optimization problem formulation in Section III-B, we show how the BiLiPD receiver is designed for a special and simple case where $B_0 = 1/T$ in Section IV-A and for general case in Subsection IV-B with the application of the task-based quantization methods. In Section IV-C, we summarize the designing procedure of the proposed BiLiPD receiver, and then, investigate the parameter estimation accuracy in BiLiPD receiver theoretically.

A. Design of the BiLiPD Receiver When $B_0 = 1/T$

In this subsection, we study the optimization of BiLiPD receiver for the special case of $B_0 = 1/T$, resulting in $N_0 = 1$. It means that during each pulse interval, each channel only outputs one digital sample. Thus, $\mathbf{B}_{m,q}$ becomes a $1 \times N$ row vector, rather than a Toeplitz matrix. Then, our objective function in (24) becomes

$$\begin{aligned} \min_{\{\mathbf{B}_1, \dots, \mathbf{B}_Q\}, \mathbf{D}, \gamma} \quad & \mathbb{E}\{\|\mathbf{y} - \hat{\mathbf{y}}\|^2\} + \lambda_C \|\Psi^H \Psi - \xi \mathbf{I}_{NQ}\|_F^2 \\ \text{s.t.} \quad & \tilde{\mathbf{B}} = \text{blkdiag}(\mathbf{B}_1, \dots, \mathbf{B}_Q) \end{aligned} \quad (25)$$

with $\mathbf{B}_q = [\mathbf{B}_{1,q}^T, \dots, \mathbf{B}_{M,q}^T]^T \in \mathbb{C}^{M \times N}$. Note that the digital filter $\mathbf{D}$ only affects the first term of the cost function in (25). We begin by characterizing the digital filter matrix that minimizes the MSE $\mathbb{E}\{\|\mathbf{y} - \hat{\mathbf{y}}\|^2\}$ for a fixed analog processing matrix $\tilde{\mathbf{B}}$. Then, we solve the problem using the gradient descent strategy.

By considering the induced quantization distortion, the optimal digital filtering $\mathbf{D}$ can be designed as an LMMSE estimator of $\mathbf{y}$. Let Σ be the covariance matrix of the received signal $\tilde{\mathbf{H}}\Phi\mathbf{a} + \mathbf{w}$. We can derive the following result.

LEMMA 1 For any analog processing matrix $\tilde{\mathbf{B}}$, if the quantizers operate within $[-\gamma, \gamma]$ with probability one, the optimal digital processing matrix that minimizes the MSE objective in (25) is given by

$$\mathbf{D}^\circ(\tilde{\mathbf{B}}) = \tilde{\mathbf{E}}\tilde{\mathbf{B}}\Sigma\tilde{\mathbf{B}}^H \left(\tilde{\mathbf{B}}\Sigma\tilde{\mathbf{B}}^H + \frac{4\gamma^2}{3b^2}\mathbf{I} \right)^{-1} \tilde{\mathbf{E}}^H \quad (26)$$

and the achievable MSE, denoted as $\epsilon_{\text{MSE}}(\tilde{\mathbf{B}}) = \min_{\mathbf{D}} \mathbb{E}\{\|\mathbf{y} - \hat{\mathbf{y}}\|_2^2\}$, is given by

$$\begin{aligned} \epsilon_{\text{MSE}}(\tilde{\mathbf{B}}) = & \text{Tr}(\tilde{\mathbf{B}}\Sigma\tilde{\mathbf{B}}^H) \\ & - \text{Tr} \left(\tilde{\mathbf{B}}\Sigma\tilde{\mathbf{B}}^H \left(\tilde{\mathbf{B}}\Sigma\tilde{\mathbf{B}}^H + \frac{4\gamma^2}{3b^2}\mathbf{I} \right)^{-1} \tilde{\mathbf{B}}\Sigma\tilde{\mathbf{B}}^H \right). \end{aligned} \quad (27)$$

PROOF The proof can be obtained from [24, Lemma 1], and thus, we omit the proof for brevity. ■

Using Lemma 1, we can optimize the analog processing matrix, which also dictates the support of the quantizers via (22). By applying [25], we can obtain the following result.

LEMMA 2 Let $\kappa = \eta^2(1 - \frac{2\eta^2}{3b^2})^{-1}$. For every analog processing matrix $\tilde{\mathbf{B}} \in \mathbb{C}^{MQ \times NQ}$, there exists a unitary matrix $\tilde{\mathbf{U}} \in \mathbb{C}^{MQ \times MQ}$ such that

$$\begin{aligned} \epsilon_{\text{MSE}}(\tilde{\mathbf{B}}) \geq & \epsilon_{\text{MSE}}(\tilde{\mathbf{U}}\tilde{\mathbf{B}}) = \text{Tr}(\tilde{\mathbf{B}}\Sigma\tilde{\mathbf{B}}^H) \\ & - \text{Tr} \left(\tilde{\mathbf{B}}\Sigma\tilde{\mathbf{B}}^H \left(\tilde{\mathbf{B}}\Sigma\tilde{\mathbf{B}}^H + \frac{4\kappa \text{Tr}(\tilde{\mathbf{B}}\Sigma\tilde{\mathbf{B}}^H)}{3b^2MQ}\mathbf{I} \right)^{-1} \tilde{\mathbf{B}}\Sigma\tilde{\mathbf{B}}^H \right). \end{aligned} \quad (28)$$

PROOF The proof is similar to that in [25, Lemma 2]. Here, we omit the proof. ■

According to the proof of Lemma 2, the minimal value of $\epsilon_{\text{MSE}}(\tilde{\mathbf{B}})$ is achieved by choosing the appropriate analog processing matrix $\tilde{\mathbf{B}}$ such that the inputs to the ADCs

have identical variance, accounting for the fact that the same quantization rule is applied to each input. Based on this observation, we rewrite the analog processing matrix optimization problem as follows:

$$\begin{aligned} \min_{\{\mathbf{B}_1, \dots, \mathbf{B}_Q\}} \quad & \underline{\mathcal{L}}(\bar{\mathbf{B}}) \\ \text{s.t.} \quad & \bar{\mathbf{B}} = \text{blkdiag}(\mathbf{B}_1, \dots, \mathbf{B}_Q) \\ & \text{Tr}(\bar{\mathbf{B}}\Sigma\bar{\mathbf{B}}^H - \delta\mathbf{I}_{MQ}) = 0 \end{aligned} \quad (29)$$

where $\delta = \frac{\text{Tr}(\Sigma)}{MQ}$ and

$$\begin{aligned} \underline{\mathcal{L}}(\bar{\mathbf{B}}) = & \lambda_C \|\Psi^H \Psi - \xi \mathbf{I}_{NQ}\|_F^2 + \text{Tr}(\bar{\mathbf{B}}\Sigma\bar{\mathbf{B}}^H) \\ & - \text{Tr}\left(\bar{\mathbf{B}}\Sigma\bar{\mathbf{B}}^H \left(\bar{\mathbf{B}}\Sigma\bar{\mathbf{B}}^H + \frac{4\kappa\delta}{3b^2}\mathbf{I}_{MQ}\right)^{-1} \bar{\mathbf{B}}\Sigma\bar{\mathbf{B}}^H\right). \end{aligned} \quad (30)$$

The second constraint in (29) ensures that each input to the ADCs possesses uniform variance, while also guaranteeing that the signal energy is unchanged after analog processing.

Problem (29) is nonconvex and challenging to solve due to the complex constraints. Let $\mathcal{L}(\bar{\mathbf{B}}) = \underline{\mathcal{L}}(\bar{\mathbf{B}}) + \lambda_T \text{Tr}(\bar{\mathbf{B}}\Sigma\bar{\mathbf{B}}^H - \delta\mathbf{I}_{MQ})$, where $\lambda_T > 0$ is the regularization parameter. Then, the problem (29) can be equivalent to the following one:

$$\begin{aligned} \bar{\mathbf{B}}^\circ = & \arg \min_{\{\mathbf{B}_1, \dots, \mathbf{B}_Q\}} \mathcal{L}(\bar{\mathbf{B}}) \\ \text{s.t.} \quad & \bar{\mathbf{B}} = \text{blkdiag}(\mathbf{B}_1, \dots, \mathbf{B}_Q). \end{aligned} \quad (31)$$

Since the function $\mathcal{L}(\bar{\mathbf{B}})$ is differentiable with respect to $\bar{\mathbf{B}}$, the optimization problem (31) can be tackled using an iterative method, which alternatively implements the gradient descent and projects the intermediate solution to a block diagonal matrix.

With the usage of multiple pulses in a CPI, the analog processing matrix $\bar{\mathbf{B}}$, denoted as $\bar{\mathbf{B}} = \text{blkdiag}(\mathbf{B}_1, \dots, \mathbf{B}_Q)$, presents block-diagonal structure, which imposes the restriction on the direct application of the gradient descent methods. If we assume that the signals between different pulses are uncorrelated with each other, the covariance matrix of the received signals Σ will also become a block diagonal matrix, i.e., $\Sigma = \text{blkdiag}(\Sigma_1, \dots, \Sigma_Q)$. Noting that the last term in (30), i.e., the Frobenius norm term $\|\Psi^H \Psi - \xi \mathbf{I}_{NQ}\|_F^2$, can be equivalently represented as

$$\begin{aligned} & \|\Psi^H \Psi - \xi \mathbf{I}_{NQ}\|_F^2 \\ &= \text{Tr}(\Psi^H \Psi \Psi^H \Psi - 2\xi \Psi^H \Psi + \xi^2 \mathbf{I}_{NQ}) \\ &= \text{Tr}(\Psi \Psi^H \Psi \Psi^H - 2\xi \Psi \Psi^H) + \xi^2 \text{Tr}(\mathbf{I}_{NQ}). \end{aligned} \quad (32)$$

The matrix $\Psi \Psi^H$, equivalent to $\bar{\mathbf{E}}\bar{\mathbf{B}}\bar{\mathbf{H}}(\bar{\mathbf{E}}\bar{\mathbf{B}}\bar{\mathbf{H}})^H$, is supposed to be a block diagonal matrix, the Frobenius norm term, i.e., $\mathbf{T}$, is thus also a block diagonal matrix, which can be denoted as $\mathbf{T} = \text{blkdiag}(\mathbf{T}_1, \dots, \mathbf{T}_Q)$. Moreover, from the standpoint of convenience, the covariance matrix of the received signals is approximated as a block diagonal matrix, i.e., $\Sigma = \text{blkdiag}(\Sigma_1, \dots, \Sigma_Q)$ to simplify optimization problem (31). We will argue the reasonableness of such approximation in the discussion part in Section IV-C.

As a result, we can break down the optimization problem defined in (31) into Q separate subproblems, each of which

Algorithm 1: Optimization of BiLiPD Receiver.

Input: $\{\mathbf{B}_q^{(0)}\}_{q=1}^Q, \{\Sigma_q\}_{q=1}^Q, K_{\max}, \lambda_C, \lambda_T, \delta, \kappa, b, \mu, \epsilon$.

Output: $\bar{\mathbf{B}}^\circ, \mathbf{D}^\circ, \gamma^\circ$.

```

1: for  $q = 1, 2, \dots, Q$  do
2:    $k \leftarrow 1, \Delta_{\mathcal{L}} \leftarrow 1$ ;
3:   Compute  $\mathcal{L}(\mathbf{B}_q^{(0)})$  according to (30);
4:   while ( $k \leq K_{\max}$  and  $\Delta_{\mathcal{L}} \geq \epsilon$ ) do
5:     Compute  $\nabla_{\mathbf{B}_q} \mathcal{L}(\mathbf{B}_q^{(k-1)})$  according to (37);
6:     Update  $\mathbf{B}_q^{(k)} \leftarrow \mathbf{B}_q^{(k-1)} - \mu \nabla_{\mathbf{B}_q} \mathcal{L}(\mathbf{B}_q^{(k-1)})$ ;
7:     Compute  $\mathcal{L}(\mathbf{B}_q^{(k)})$  according to (30);
8:     Update  $\Delta_{\mathcal{L}} \leftarrow \mathcal{L}(\mathbf{B}_q^{(k)}) - \mathcal{L}(\mathbf{B}_q^{(k-1)})$ ;
9:      $k \leftarrow k + 1$ ;
10:  end while
11: end for
12: return  $\bar{\mathbf{B}}^\circ$ ;
13: Compute  $\gamma^\circ$  and  $\mathbf{D}^\circ$  according to (22) and (26).
```

can be tackled independently using gradient descent methods. The q th optimization problem is expressed as follows

$$\mathbf{B}_q^\circ = \arg \min_{\mathbf{B}_q} \mathcal{L}(\mathbf{B}_q). \quad (33)$$

Let the derivative of the cost function $\mathcal{L}(\mathbf{B}_q)$ with respect to $\mathbf{B}_q$ be denoted as $\nabla_{\mathbf{B}_q} \mathcal{L}(\mathbf{B}_q)$, whose mathematical expression is provided in Appendix. In accordance with the gradient descent methodology, during each iteration, the analog processing matrix $\mathbf{B}_q$ is updated. The specific procedure is outlined as follows.

B. Design of the BiLiPD Receiver for General Case

In the more general scenario where $N_0 > 1$, addressing the problem (24) directly presents challenges due to the complex nature of the analog matrix $\bar{\mathbf{B}}$, which possesses both a block-diagonal and a Toeplitz structure. One approach to solve (24) is to first relax it to a problem, disregarding the Toeplitz structure of $\bar{\mathbf{B}}$. Following this relaxation, once we have derived the optimal solution $\bar{\mathbf{B}}^*$, we proceed to define the desired $\bar{\mathbf{B}}$ by minimizing the discrepancy between each element in $\bar{\mathbf{B}}$ and its corresponding value in $\bar{\mathbf{B}}^*$. This results in the following optimization problem:

$$\begin{aligned} \min_{\{\mathbf{B}_{m,q}\}} \quad & \|\bar{\mathbf{B}} - \bar{\mathbf{B}}^*\|_F^2 \\ \text{s.t.} \quad & \bar{\mathbf{B}} = \text{blkdiag}(\mathbf{B}_1, \dots, \mathbf{B}_Q) \\ & \mathbf{B}_{m,q} \in \mathbb{T}^{N_0 \times N} \\ & m = 1, 2, \dots, M, q = 0, 1, \dots, Q-1 \end{aligned} \quad (34)$$

where $\mathbf{B}_q = [\mathbf{B}_{1,q}^T, \dots, \mathbf{B}_{M,q}^T]^T$, and the Frobenius norm, defined as $\|\mathbf{A}\|_F^2 = \sum_{i=1}^m \sum_{j=1}^n |a_{i,j}|^2$ for $\mathbf{A} \in \mathbb{C}^{m \times n}$, is used to quantify the distance between the desired $\bar{\mathbf{B}}$ and the solution $\bar{\mathbf{B}}^*$.

To clarify the process, we first obtain the unconstrained matrix $\bar{\mathbf{B}}^*$, denoted as $\bar{\mathbf{B}}^* = \text{blkdiag}(\mathbf{B}_1^*, \dots, \mathbf{B}_Q^*)$, by following the procedure outlined in Section IV-A. Subsequently, we seek the minimal-norm solution that bridges

the gap between the unconstrained $\bar{\mathbf{B}}^*$ and our desired $\bar{\mathbf{B}}$. It is worth noting that within the BiLiPD receiver architecture, the mixing of signals across different channels during each pulse is independent. Therefore, each submatrix $\mathbf{B}_{m,q}$ can be individually designed, and the optimization problem for each submatrix is expressed as follows:

$$\begin{aligned} \min_{\{\mathbf{B}_{m,q}\}} & \|\mathbf{B}_{m,q} - \mathbf{B}_{m,q}^*\|_F^2 \\ \text{s.t. } & \mathbf{B}_{m,q} \in \mathbb{T}^{N_0 \times N} \\ & m = 1, 2, \dots, M, q = 0, 1, \dots, Q - 1. \end{aligned} \quad (35)$$

The aforementioned problem can be explicitly solved by applying the Toeplitz operator $\mathcal{T}(\cdot)$, which projects a matrix into a Toeplitz matrix, i.e., the solution is $\mathbf{B}_{m,q}^\circ = \mathcal{T}(\mathbf{B}_{m,q}^*)$. According to the structure of Toeplitz matrix in which each descending diagonal element from left to right is constant, the explicit closed-form solution of $\mathbf{B}_{m,q}^\circ$ can also be directly given by setting the elements of the desired Toeplitz matrix $\mathbf{B}_{m,q}^\circ$ that are located on the line parallel to the main diagonal, to be the same with their values equal to the average value of the elements that located on the corresponding line in the unconstrained $\mathbf{B}_{m,q}^*$. Once the optimized Toeplitz matrix $\mathbf{B}_{m,q}^\circ$ is calculated, we obtain the desired analog combining matrix $\bar{\mathbf{B}}^\circ = \text{blkdiag}(\mathbf{B}_1^\circ, \dots, \mathbf{B}_Q^\circ)$ with $\mathbf{B}_q^\circ$ denoted as $\mathbf{B}_q^\circ = [(\mathbf{B}_{1,q}^\circ)^T, \dots, (\mathbf{B}_{M,q}^\circ)^T]^T$. Then, the digital filter $\mathbf{D}$ and dynamic range of quantizers γ can be calculated according to (22) and (26), respectively.

C. Discussion

In our proposed BiLiPD receiver, we address the joint challenges posed by sub-Nyquist sampling and low-bit quantization. We formulate this intricate design problem as (24). Our system is designed to acquire sub-Nyquist samples with low-coherence measurements, a key factor contributing to robust sparse recovery performance. Simultaneously, we implement a design that focuses on minimizing quantization distortion through the utilization of the task-based quantization framework. The outcome is achieved by carefully adjusting the analog processing matrix $\bar{\mathbf{B}}$, the dynamic range of quantizers γ , and the digital filter $\mathbf{D}$.

In contrast to prior works [25], our approach configures the sub-Nyquist samples $\mathbf{y}$ as the IT vector. The whole system is then fine-tuned to produce an output vector $\hat{\mathbf{y}}$, which is served as an accurate estimate of $\mathbf{y}$. This precision is crucial, as it enables the accurate recovery of target information from $\hat{\mathbf{y}}$. By incorporating the task-based quantization framework and leveraging the mutual coherence of the measurement matrix, the building modules of the BiLiPD receiver undergo a holistic optimization process. This comprehensive approach is aimed at mitigating the effects of quantization distortion while enhancing the performance of sparse recovery.

The proposed BiLiPD receiver shares a similar architecture with the Xampling framework discussed in [32], which employs a combination of modulators and low-pass filters to perform analog preprocessing. This architecture

serves a dual purpose: it enables sub-Nyquist sampling and also provides a controllable analog hardware platform for implementing analog preprocessing tailored to task-based quantization requirements. Consequently, our BiLiPD architecture offers a concrete and practical design framework within the context of task-based quantization. In contrast, prior works in [24] primarily relied on the assumption of a linear analog map during their designs, lacking the explicit architecture needed to realize controllable analog hardware.

In this work, the prior knowledge of the underlying statistical model, specifically, the covariance matrix Σ , is exploited to facilitate the optimization of the acquisition system as is given in (30). It is noted that the actual response of the environment is actually unavailable before hand. We denote $\mathbf{R}_c$ and $\mathbf{R}_w$ as the covariance matrices of the pure signal $\bar{\mathbf{H}}\Phi\mathbf{a}$ and noise signal $\mathbf{w}$, respectively. In this work, the estimation of the pure signal covariance matrix $\mathbf{R}_c$ is built with the assumption that the targets are distributed randomly in the delay-Doppler region. The prior knowledge of the statistic model of noise, i.e., $\mathbf{R}_w$, is supposed to be commonly available in many scenes. Consequently, the covariance matrix of received signal used in optimization procedure, denoted as $\Sigma = \mathbf{R}_c + \mathbf{R}_w$, is just the coarse estimation of the real one without any prior knowledge of target distribution. Note that such the coarse estimation Σ leads to imperfect optimization of the system design and more accurate knowledge of the signal statistic model is supposed to improve the overall performance. In radar tracking scenarios, the accurate knowledge of the signal statistic model can be obtained from previous observations. An alternative approach is to obtain the information involves collecting data and estimating the covariance matrix during the warm-up processing phase, where we apply an initial setup to the BiLiPD receiver. Developing efficient strategies for acquiring the covariance matrix will be our future work.

In previous subsection, the covariance matrix Σ is approximated as a block diagonal matrix from the standpoint of convenience so that the optimization problem can be broken down into Q individual subproblems. Such approximation is supposed to be reasonable from two aspects. For one thing, Σ itself is the coarse estimation of the actual response of environment signal. For another, the coarse estimation obtained with the assumption that the targets are distributed randomly in the delay-Doppler region, leading to weak coherence between across pulses.

V. NUMERICAL RESULTS

In this section, we perform the numerical evaluation of the proposed BiLiPD receiver. We compare our method against two benchmarks: the classic PD radar processing method with unquantized data and the PD radar processing with task-ignorant quantizers. It is worth noting that the design of the BiLiPD receiver incorporates task-based quantization methods, specifically aimed at obtaining an accurate estimate of IT vector $\mathbf{y}$. This vector, in turn, allows for the resolution of target parameters embedded within the

sparse vector $\mathbf{a}$ through the use of the FISTA algorithm to solve the LASSO problem (19).

A. Simulation Setup

We consider a PD radar using LFM signal with the signal bandwidth $B = 6$ MHz and pulsewidth $T_p = 1\mu s$. The remaining simulation parameters are: the number of transmitted pulses in a CPI $Q = 64$, PRI $T = 20\mu s$, and carrier frequency $f_c = 10$ GHz. The number of samples during a pulse, denoted as $N = BT$, can be derived as $N = 120$ with the sampling rate set to be equivalent to the bandwidth B .

We consider $K = 4$ targets with the target model assumption given in Section II-A, and the targets reflection coefficients $\{\alpha_k\}$ are generated according to the Swerling 1 model. Consider that the parameters τ_k and f_k of each target are generated on the delay-Doppler grids as defined in Section II-C with grid spacing given as $\Delta_\tau = \frac{T}{N}$ and $\Delta_f = \frac{1}{QT}$, respectively, i.e., $\tau_k \in \{0, \Delta_\tau, 2\Delta_\tau, \dots, T\}$, $f_k \in \{-\frac{1}{2T}, -\frac{1}{2T} + \Delta_f, -\frac{1}{2T} + 2\Delta_f, \dots, \frac{1}{2T}\}$; and that τ_k and f_k remain unchanged during one CPI. The i.i.d. additive proper-complex Gaussian noise model is assumed that noise with zero mean and variance σ_n^2 between different pulses is uncorrelated. Accordingly, the signal-to-noise ratio (SNR) is defined as $\text{SNR} = \frac{\mathbb{E}(\|\mathbf{H}\Phi\mathbf{a}\|_2^2)}{NK\sigma_n^2}$.

In the proposed BiLiPD receiver, we consider M channels in the analog processing with the bandwidth of low-pass filters denoted as B_0 . The number of samples is reduced to MN_0Q via the analog combining and Xampling processing. The compression ratio Δ_{cr} , denoted to as $\Delta_{cr} = \frac{N}{MN_0Q}$, is introduced to indicate the degree of the compression between the desired parameter vector $\mathbf{a}$ and the compressed vector $\mathbf{y}$. The specific values of simulation parameters are given before each experiment.

We compare the performance of the BiLiPD receiver with two different systems or recovery methods. The one that quantizes the received signals regardless of the system task under the same overall bit-budget constraints as in BiLiPD receiver, is referred to as “Task-ignorant Quan.” method. In such PD radar receiver with task-ignorant quantizers, the target parameters are resolved directly from the quantized outputs without analog compression and digital processing. The PD radar without quantization, referred to as “No Quan.” method, recovers the target parameter vector $\mathbf{a}$ directly from the received signal $x(t)$. Note that the “No Quan.” method is actually unavailable in reality. For the proposed BiLiPD receiver, three different parameter settings are investigated. Referring to the Section IV-A, the regularization parameters, i.e., λ_C and λ_T , are essential to the designing of the BiLiPD receiver. The first BiLiPD receiver, denoted as “BiLiPD.1” method, is designed in consideration of both MSE in IT vector $\mathbf{y}$ and incoherence property of the equivalent matrix Ψ by setting $\lambda_T = 0.2$ and $\lambda_C = 0.3$. The one that only focuses on minimizing the MSE in $\mathbf{y}$ by setting $\lambda_T = 0.2$ and $\lambda_C = 0$, is denoted as “BiLiPD.2.” For comparison, the “BiLiPD.3” method is proposed by setting the analog matrix $\tilde{\mathbf{B}}$ as a complex

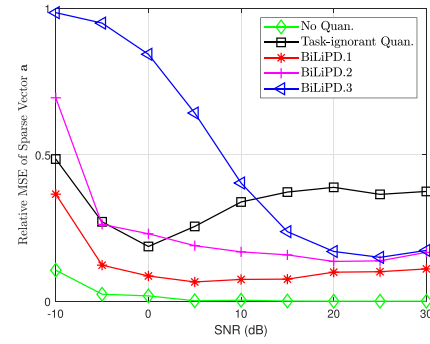


Fig. 3. MSE in recovering $\mathbf{a}$ versus SNR ($K = 4$).

Gaussian random matrix, which is supposed to achieve great incoherence property of the equivalent matrix Ψ . Throughout the simulations, the performance of these various methods is evaluated numerically by carrying out Monte Carlo experiment over 100 realizations.

B. Recovery Performance

We evaluate the recovery performance of these methods by the following two metrics.

- 1) **Relative MSEs of target vector $\mathbf{a}$:** The relative MSEs of target vector $\mathbf{a}$ with the target parameters embedded in, are defined as $\|\mathbf{a} - \hat{\mathbf{a}}\|_2^2 / \|\mathbf{a}\|_2^2$.
- 1) **Hit rate:** A “Hit” is achieved when the estimated target vector $\hat{\mathbf{a}}$ is exactly equal to the true target vector $\mathbf{a}$, given that the target parameters are defined to fall on the grids.

The aforementioned two metrics are utilized to evaluate the target recovery performance, which are considered to be the actual goals of the radar receiver systems. Specifically, the relative MSE of target vector $\mathbf{a}$ illustrates the estimation accuracy of the targets’ inflection amplitudes $\{\alpha_k\}_{k=1}^K$; while the latter one, which is referred to as hit rate, is utilized to illustrate the estimation accuracy of range Doppler pairs, i.e., $\{\tau_k, f_k\}_{k=1}^K$. Specifically, in the investigation of “hit rate” performance, the amplitude of the reflection coefficients of each target is fixed to unity while its phase is randomly distributed.

We first evaluate the recovery performance of these mentioned recovery methods versus the SNR. The performance of three different BiLiPD receivers are investigated for comparison. The total number of bits is fixed to $2QN = 15360$, leading to 1-bit quantization scheme for the task-ignorant method with two bits per complex sample. We set the compression ratio of BiLiPD receivers as $\Delta_{cr} = 4$, the number of channels $M = 10$, the bandwidth of low-pass filters $B_0 = 0.3$ MHz, and the number of samples $MN_0Q = 1920$.

Fig. 3 illustrates the relative MSE in recovering the sparse vector $\mathbf{a}$ with respect to the SNR. It is observed that the “No Quan.” method achieves the best estimation performance with its MSE approaching to zero in SNRs

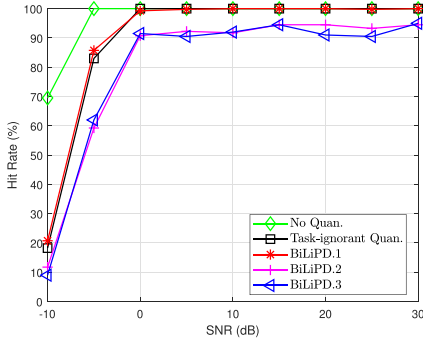


Fig. 4. Hit rate versus SNR ($K = 4$).

beyond 0 dB and the estimation performance of the “Task-ignorant Quan.” method is unsatisfactory owing to the limited bit budgets. Compared the performance between three different BiLiPD receivers, it is evident that the “BiLiPD.2” and “BiLiPD.3” both perform worse than the proposed “BiLiPD.1” receiver, which takes both MSE of IT vector and incoherence performance of measurement matrix into consideration. Moreover, with the increasing SNR, the performance improvement of “BiLiPD.3” is not significant as it does not take the MSE of IT vector into consideration. It seems strange to see an uptrend of the estimation error even with the increase of SNR in the “Task-ignorant Quan.” curve for SNRs beyond 0 dB. Such curious trend arises from the characteristic of the 1-bit quantization scheme that the 1-bit quantization is a highly nonlinear system and noises may be helpful in this kind of nonlinear systems, allowing for some recovery of the amplitude of sparse vector $\mathbf{a}$ [36].

When it comes to the hit rate versus the SNR as depicted in Fig. 4, it is evident that the “No Quan.” method outperforms all other recovery methods. In addition, both the proposed “BiLiPD.1” method and the “Task-ignorant Quan.” method achieve reliable parameter estimation performance with “hit rate” approaching to 100% for SNRs beyond 0 dB. However, the other two “BiLiPD” receivers, i.e., the “BiLiPD.2” and “BiLiPD.3” methods fail to obtain satisfactory performance even in the high SNR regime as they only focus on one single aspect of optimization according to (30).

To better find out the effect of the compression ratios on the performance, we now investigate the recovery performance of the BiLiPD receiver under different compression ratios with a fixed number of channels $M = 10$. In this simulation, we adopt the “BiLiPD.1” method, which is labeled as “BiLiPD.” in the following. The total number of bits is fixed to $2QN = 15360$. We set four different compression ratios for BiLiPD receivers and compare their performance. The corresponding parameters, i.e., the compression ratios Δ_{cr} , the number of channels M , the bandwidth of low-pass filters B_0 , and the number of samples MN_0Q , are given in Table I.

As is shown in Fig. 5, the BiLiPD receivers with $\Delta_{cr} = 4$ and $\Delta_{cr} = 6$ both outperform the other two receivers. Specifically, the receiver with $\Delta_{cr} = 4$ demonstrates a slight advantage over the receiver with $\Delta_{cr} = 6$, and

TABLE I
Parameters Setting Under Different
Compression Ratios

Δ_{cr}	M	B_0	MN_0Q
2	10	150KHz	3840
4	10	75KHz	1920
6	10	50MHz	1280
12	10	25MHz	640

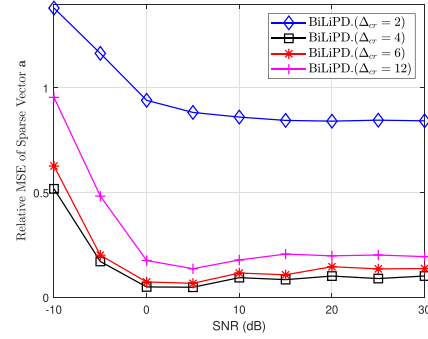


Fig. 5. MSE in recovering $\mathbf{a}$ under different compression ratios ($M = 10$).

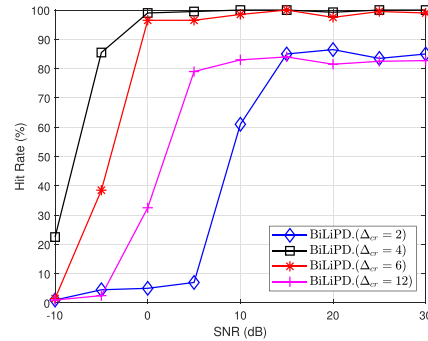


Fig. 6. Hit rate under different SNR compression ratios ($M = 10$).

both receivers achieve satisfactory performance for SNRs beyond 0 dB. However, the “BiLiPD. ($\Delta_{cr} = 2$)” receiver fails to achieve the desired performance compared to the other receivers. At low SNRs, the noise is the main source of estimation error, making the performance noise-limited. As the SNR increases, quantization error gradually becomes the dominant source of error, making the performance quantization-limited. In CS literature, the compression of the measurements has a substantial impact on SNR, degrading recovery performance due to a phenomenon known as noise folding. Although the compression in BiLiPD receivers save more bits per sample, the noise folding induced by compressed sampling degrades estimation performance, which is particularly obvious at low SNRs.

Fig. 6 illustrates the hit rate performance of BiLiPD receivers under different compression ratios versus the SNR. It is observed that the BiLiPD receivers with $\Delta_{cr} = 12$ and $\Delta_{cr} = 2$ achieve worse performance compared to the other two BiLiPD receivers. By comparing the two BiLiPD receivers with $\Delta_{cr} = 4$ and $\Delta_{cr} = 6$, we note that the BiLiPD receiver with $\Delta_{cr} = 4$ performs better than the other one at

TABLE II
Parameters Setting Under Different
Compression Ratios

Δ_{cr}	M	B_0	MN_0Q
4	3	0.5MHz	1920
4	5	0.3MHz	1920
4	6	0.25MHz	1920
4	10	0.15MHz	1920

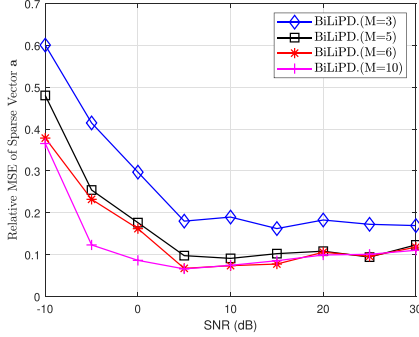


Fig. 7. MSE in recovering $\mathbf{a}$ with different number of channels ($\Delta_{cr} = 4$).

low SNR, while the BiLiPD receiver with $\Delta_{cr} = 6$ achieves equally satisfying result as receiver with $\Delta_{cr} = 4$ in a high SNR regime.

The BiLiPD receivers with different compression ratios exhibit varying performance levels. We note that the selection of an appropriate compression ratio is vital to the BiLiPD design as both inadequate compression as well as excessive compression can hinder the successful recovery of the sparse vector. A reasonable compression ratio is supposed to make the balance between the quantization distortion and the noise folding effects. Therefore, there exists a tradeoff between reducing quantization distortion and maintaining good sparse recovery performance when selecting the compression ratio for the BiLiPD receiver.

Next, we fix the compression ratio to $\Delta_{cr} = 4$ and investigate the recovery performance of the BiLiPD receivers with different number of channels M . Given a fixed compression ratio, the number of channels in the BiLiPD receiver is adjusted by dynamically changing the bandwidth of the low-pass filters denoted as B_0 . Specifically, a larger number of channels can be achieved by reducing the value of B_0 , while ensuring that the total number of samples remains unchanged. The specific parameters are given in Table II.

It is noticed in Fig. 7 that the BiLiPD receiver with more channels consistently outperforms those with fewer channels. The estimation errors of the BiLiPD receivers with more than five channels decrease to nearly 0.1 for SNRs beyond 5 dB. It is noted that during the q th pulse, the submatrices $\{\mathbf{B}_{m,q}\}_{m=1}^M \in \mathbb{T}^{N_0 \times N}$ are constrained by the Toeplitz structure according to practical mixing operation as is given in (10). The projection of the optimal solutions $\{\mathbf{B}_{m,q}^*\}_{m=1}^M$ onto the Toeplitz matrices deforms the theoretically optimal solutions in (33) following our optimization procedure in Section IV-B, which is supposed to subsequently degrade the recovery performance of the proposed BiLiPD receivers.

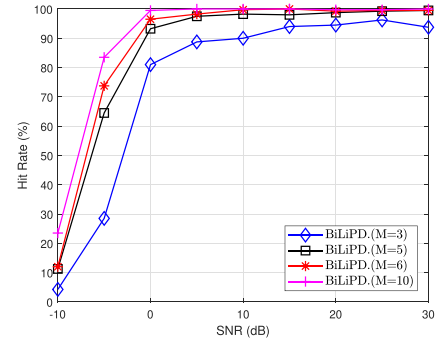


Fig. 8. Hit rate with different number of channels ($\Delta_{cr} = 4$).

As the projection is conducted channel-wise, the system with more channels, which corresponds to smaller dimensional $\{\mathbf{B}_{m,q}\}_{m=1}^M$ with a fixed MN_0 , provides more degrees of freedom during the projection operation, and is supposed to be less influenced from projection deformation. In other words, when dealing with matrices with larger dimensions, all entries in $\mathbf{B}_{m,q}^*$ need to be modified to conform to the Toeplitz form required by the structural constraints, which suffer more from the matrix deformation. Consequently, the BiLiPD receiver with fewer channels, which corresponds to a higher dimensional $\{\mathbf{B}_{m,q}\}_{m=1}^M$, tends to experience greater impact from the projection step shown in (35), thus suffering more from significant degradation of the recovery performance.

The results depicted in Fig. 8 are entirely consistent with that in estimation performance, demonstrating that BiLiPD receivers with more channels consistently outperform those with fewer channels under the same compression ratio. In contrast to the results seen in Figs. 5 and 6, which suggest the existence of an optimized value for the compression ratio Δ_{cr} , it seems that the BiLiPD receivers tend to benefit from having as many channels as possible. Indeed, the performance improvement gained by increasing the number of channels comes at the expense of additional RF chains and ADC devices, which increases the burden of hardware cost. Moreover, the increasing number of channels brings about the difficulty in time synchronization between different channels. Overall, the tradeoff between performance enhancement and hardware complexity needs to be carefully considered when choosing the number of channels M .

C. Practical Concerns

At the end part, we discuss some practical concerns, which are generally ignored in conventional sub-Nyquist radars.

1) *Dynamic Range Problem:* In real-world conditions, a strong target can obscure a weak target, referred to as the dynamic range problem. We here investigate the dynamic range problem of the bit-limited receivers for detecting two targets with different amplitudes. In this simulation, we fix the amplitude of one target as 1, and change the amplitude of the other target. The target amplitude ratio index, which is defined to measure the different amplitudes between these two targets, varies from 2^0 to 2^9 on a logarithmic scale.

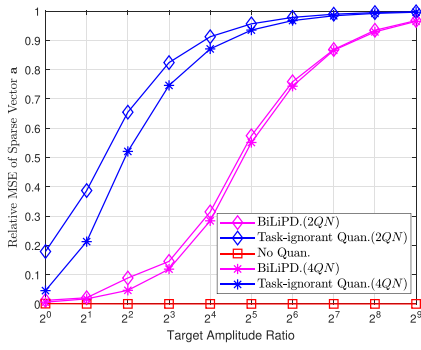


Fig. 9. MSE in recovering $\mathbf{a}$ versus target amplitude ratio under different bit budgets ($K = 2$).

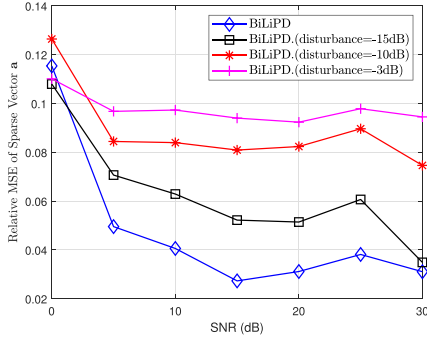


Fig. 10. MSE in recovering $\mathbf{a}$ versus SNR with inaccurate analog processing matrices.

Fig. 9 shows the relative MSE of recovering the sparse vector $\mathbf{a}$ versus the target amplitude ratio. It is demonstrated that the BiLiPD receiver is capable of recovering the weak target's parameters when the amplitude ratio is small, while the MSE performance severely degrades as the amplitude ratio increases. Both the limited bit-budget systems, i.e., the "BiLiPD" receiver and "Task-ignorant" receiver, suffer from dynamic range problems with estimation quality degrading significantly due to quantization. With the increasing total number of bits, the quantization errors decrease, thereby mitigating the estimation performance degradation of both limited bit-budget systems. To solve the dynamic range problem of low-bit ADCs, a mixed-ADC-based architecture [37] may be applied.

2) *Effect of Hardware Nonidealities*: We now evaluate the effect of hardware nonidealities on the BiLiPD receiver's recovery performance. Since the implementation of accurate analog preprocessing is a significant challenge in practical applications, we conduct the simulation experiments by adding some disturbance on the designed analog preprocessing matrix $\bar{\mathbf{B}}$, which models the hardware nonidealities. We define the level of disturbance as $\text{SNR}_d = \|\mathbf{N}\|_2^2 / \|\bar{\mathbf{B}}\|_2^2$, where $\mathbf{N}$ denotes the disturbance matrix.

Fig. 10 shows the relative MSE in recovering the sparse vector $\mathbf{a}$ versus SNR with different disturbance levels. It is observed that the hardware nonidealities of the analog preprocessing module will degrade the recovery performance. Fortunately, when the disturbance is small, the performance

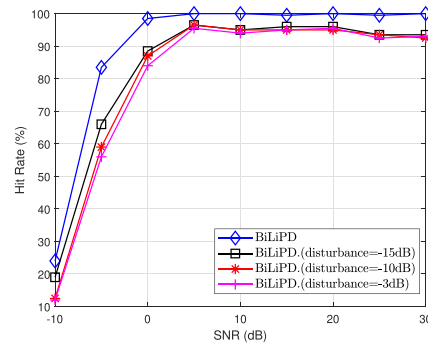


Fig. 11. Hit rate versus SNR with inaccurate analog processing matrices.

degradation is acceptable. However, as the disturbance increases, the MSE performance deteriorates significantly.

Fig. 11 shows the hit rate curves of the BiLiPD receiver versus the SNR with different disturbance levels. The performance degradation is also observable due to the hardware nonidealities. In particular, it is unable to reach 100% hit rate even at high SNR scenarios.

These results show that the proposed BiLiPD receiver can bear minor disturbance due to hardware nonidealities with slight performance degradation. However, the performance loss will increase when the analog hardware significantly deviates from the ideal design. Therefore, the robust design of the BiLiPD receiver to deal with hardware nonidealities will be our future work.

3) *Receiver Operating Characteristic*: In CS radar applications, the estimation of targets is typically modeled as the reconstruction of sparse vectors. The solution of sparse reconstruction lacks analytical expressions and clear statistical properties, which complicates the analysis of detection performance. However, assessing the detection probability and controlling the false alarm rate is critical in practical scenarios. Several works [38], [39], [40] have explored the use of debiased LASSO to design CS radar detectors, as it enables the formulation of test statistics. Nonetheless, these methods generally assume Gaussian, row-orthogonal, or random discrete cosine transformation measurement matrices, which do not align well with our model. In this article, we apply the debiased LASSO to the outputs of the BiLiPD receiver. Then, we use the residual sum of squares, defined as $\text{RSS} = \frac{1}{N} \|\mathbf{y} - \mathbf{A}\mathbf{x}^{\text{LASSO}}\|_2^2$ in [40], to design a detector for demonstrating the empirical detection performance of the BiLiPD receiver.

Figs. 12 and 13 show the curves of the detection probability and false alarm rate, respectively. It is shown that the detection probability curves for all receivers rise sharply with increasing SNR while maintaining false alarm rates within acceptable limits. Specifically, the "BiLiPD.1" receiver achieves the best detection performance among all the bit-limited receivers, approaching the performance of the "No Quan" receiver. The derivation of the theoretical detection threshold for designing a constant false alarm detector represents a promising area of research for the future.

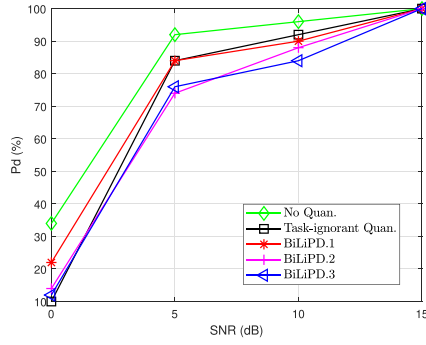


Fig. 12. Detection probability versus SNR with a varying threshold.

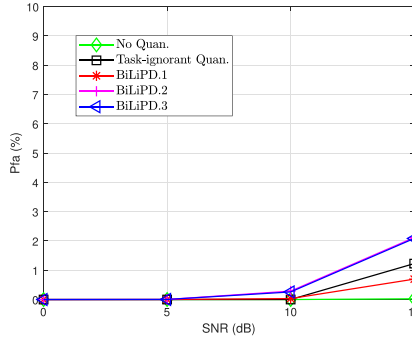


Fig. 13. False alarm rate versus SNR with a varying threshold.

VI. CONCLUSION

In this work, we proposed a BiLiPD radar receiver with the combination of sub-Nyquist sampling framework and the task-based quantization methods. We applied the HAD architecture to the PD radar receiver by jointly optimizing the analog component, the low-rate ADCs and the digital processing component with the bit budget limitation. Particularly, the analog component in the BiLiPD receiver was modulated for practical applications under the structure constraints. The simulation results validate the effectiveness of the proposed BiLiPD receiver that the BiLiPD receiver achieves better performance compared to the conventional processing method with the same bit budget.

APPENDIX

We present the gradient descent procedure explicitly. As the optimization of analog processing during different pulses can be achieved individually, we rewrite the optimization function of the q th pulse as

$$\begin{aligned} \mathcal{L}(\mathbf{B}_q)^* &= \text{Tr}(\mathbf{B}_q \boldsymbol{\Sigma}_q \mathbf{B}_q^H - \mathbf{B}_q \boldsymbol{\Sigma}_q \mathbf{B}_q^H (\mathbf{B}_q \boldsymbol{\Sigma}_q \mathbf{B}_q^H \\ &\quad + \frac{4\eta^2 \text{Tr}(\mathbf{B}_q \boldsymbol{\Sigma}_q \mathbf{B}_q^H)}{3b^2 MN_0} \mathbf{I})^{-1} \mathbf{B}_q \boldsymbol{\Sigma}_q \mathbf{B}_q^H) / \text{Tr}(\mathbf{B}_q \boldsymbol{\Sigma}_q \mathbf{B}_q^H) \\ &\quad (a) \\ &\quad + \lambda_T \text{Tr}(\mathbf{B}_q \boldsymbol{\Sigma}_q \mathbf{B}_q^H - \delta \mathbf{I}_{MN_0})^2 / \text{Tr}(\mathbf{B}_q \boldsymbol{\Sigma}_q \mathbf{B}_q^H)^2 \\ &\quad (b) \\ &\quad + \lambda_C \|\mathbf{M}_q^H \mathbf{M}_q - \xi \mathbf{I}_N\|_F^2 / \|\mathbf{M}_q^H \mathbf{M}_q\|_F^2 \end{aligned} \quad (36) \quad (c)$$

with $\mathbf{M}_q = \Phi_q \mathbf{H} \mathbf{B}_q$. The numerator terms in (36) are used to restrict the normalized errors of the cost function.

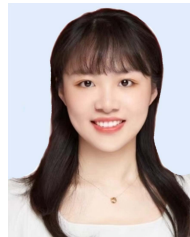
Specifically, the gradient derivation of the (a) term in (36) is given as $\nabla_{\mathbf{B}_q} \mathcal{L}^{(a)} = 2(\mathbf{B}_q \boldsymbol{\Sigma}_q - \mathbf{U}_q \mathbf{V}_q \mathbf{B}_q \boldsymbol{\Sigma}_q - \mathbf{V}_q \mathbf{U}_q \mathbf{B}_q \boldsymbol{\Sigma}_q + (1 + \frac{4\eta^2}{3b^2 MN_0}) \mathbf{V}_q \mathbf{U}_q \mathbf{U}_q \mathbf{V}_q \mathbf{B}_q \boldsymbol{\Sigma}_q) / \text{Tr}(\mathbf{U}_q) - 2 \text{Tr}(\mathbf{U}_q - \mathbf{U}_q \mathbf{V}_q \mathbf{U}_q) / \text{Tr}(\mathbf{U}_q)^2 \cdot \mathbf{B}_q \boldsymbol{\Sigma}_q$, where $\mathbf{U}_q \triangleq \mathbf{B}_q \boldsymbol{\Sigma}_q \mathbf{B}_q^H$ and $\mathbf{V}_q \triangleq (\mathbf{B}_q \boldsymbol{\Sigma}_q \mathbf{B}_q^H + \frac{4\eta^2 \text{Tr}(\mathbf{B}_q \boldsymbol{\Sigma}_q \mathbf{B}_q^H)}{3b^2 MN_0} \mathbf{I})^{-1}$. Similarly, the gradient derivation of the (b) term is given as $\nabla_{\mathbf{B}_q} \mathcal{L}^{(b)} = 4(\text{Tr}(\mathbf{U}_q - \delta \mathbf{I}_{MN_0}) / \text{Tr}(\mathbf{U}_q)^2 - \text{Tr}(\mathbf{U}_q - \delta \mathbf{I}_{MN_0})^2 / \text{Tr}(\mathbf{U}_q)^3) \mathbf{B}_q \boldsymbol{\Sigma}_q$. For the (c) term, the parameter ξ is calculated before each iteration, given as $\xi = \text{Tr}(\mathbf{M}_q^H \mathbf{M}_q) / N$. Then, the gradient derivation of the (c) term is given as $\nabla_{\mathbf{B}_q} \mathcal{L}^{(c)} = 4(\mathbf{M}_q \mathbf{M}_q^H \mathbf{M}_q \Phi_q^H \mathbf{H}^H - 2\xi \mathbf{M}_q \Phi_q^H \mathbf{H}^H) / \text{Tr}(\mathbf{M}_q \mathbf{M}_q^H \mathbf{M}_q \Phi_q^H \mathbf{H}^H) - 4\|\mathbf{M}_q^H \mathbf{M}_q - \xi \mathbf{I}_N\|_F^2 / \|\mathbf{M}_q^H \mathbf{M}_q\|_F^4 \cdot \mathbf{M}_q \mathbf{M}_q^H \mathbf{M}_q \Phi_q^H \mathbf{H}^H$. In summary, the gradient of the objective function (36) of the q th pulse can be presented as

$$\nabla_{\mathbf{B}_q} \mathcal{L}(\mathbf{B}_q)^* = \nabla_{\mathbf{B}_q} \mathcal{L}^{(a)} + \lambda_T \nabla_{\mathbf{B}_q} \mathcal{L}^{(b)} + \lambda_C \nabla_{\mathbf{B}_q} \mathcal{L}^{(c)}. \quad (37)$$

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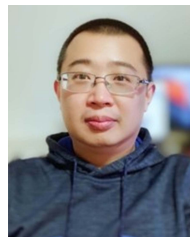
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